



BIG DATA

2025/26

Ciclo	Especialización IA & BIG DATA
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PR_04.2

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1. APARTADO A

1. Cambiar contraseña de root de MySQL

Al ejecutar el siguiente comando, la configuración por defecto de la autenticación de root queda establecida como *auth_socket*, permitiendo acceder como root sin contraseña y solo a través del localhost

```
sudo mysql_secure_installation
```

Si accedemos a mysql como root, ahora podremos cambiar la contraseña con:

```
ALTER USER 'root'@'localhost' IDENTIFIED WITH  
mysql_native_password BY 'nuevaContrasena';
```

Para forzar a root a autenticarse con su contraseña establecida y permitir acceso remoto a la bbdd, más allá del acceso en localhost

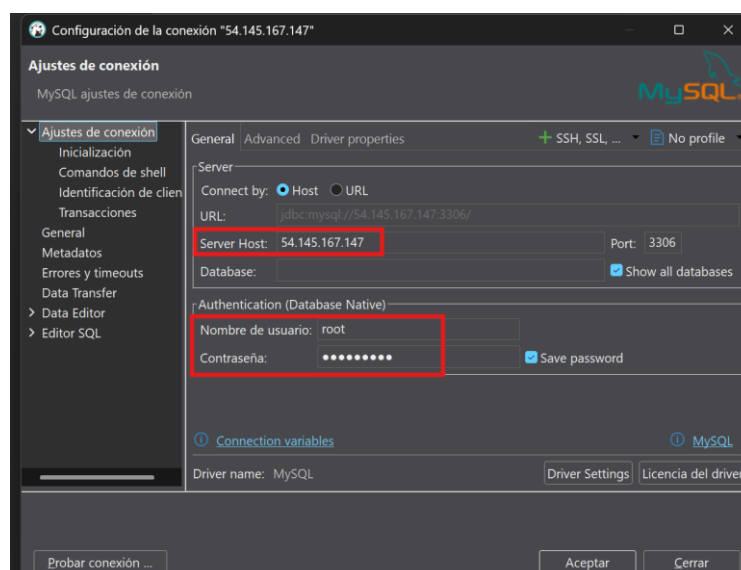
localhost → % (acceso remoto desde cualquier host)

```
RENAME USER 'root'@'localhost' TO 'root'@'%';
```

Aplicamos cambios

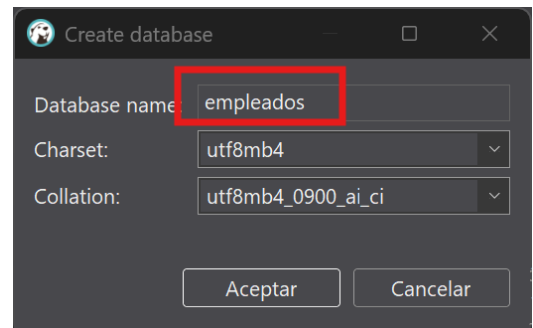
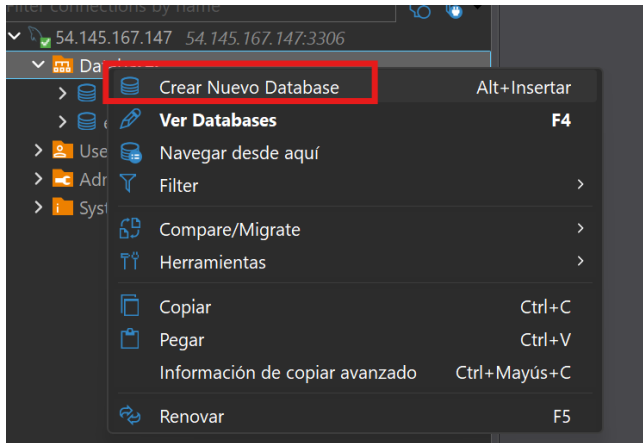
```
FLUSH PRIVILEGES;
```

2. Conectarse con DBeaver al servidor MySQL.Normal



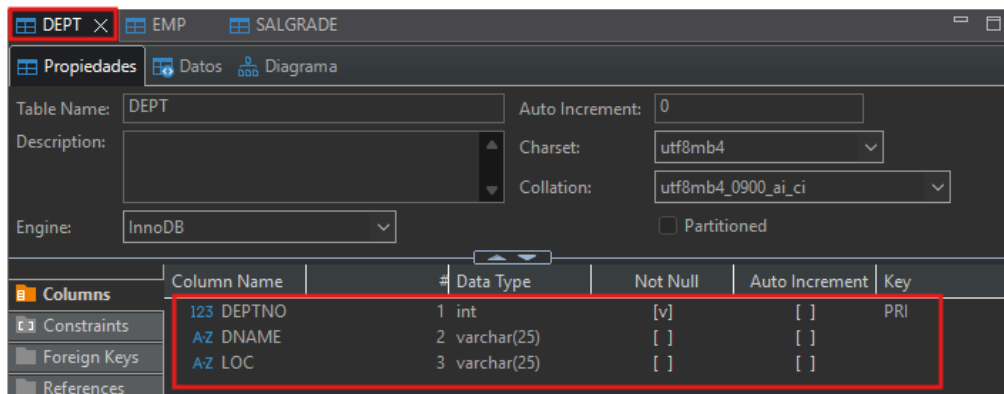
2. APARTADO B

1. Crea desde DBeaver en tu servidor MySQL una base de datos llamada **empleados**



2. Dentro de la base de datos anterior crea las tablas con los campos que se ven abajo y relaciones adecuadas entre las tablas:

DEPARTAMENTOS



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EMPLEADOS

The screenshot shows the MySQL Workbench interface for the EMP table. The 'EMP' tab is selected. The 'Propiedades' tab is active, showing the table name 'EMP', engine 'InnoDB', auto increment '0', and charset 'utf8mb4'. The 'Columns' tab is also visible, showing a list of columns with their data types and constraints. The 'Foreign Keys' tab is selected, showing two foreign keys: EMP_DEPT_FK and EMP_EMP_FK.

Column Name	#	Data Type	Not Null	Auto Increment	Key
123 ENO	1	int	[v]	[]	PRI
AZ ENAME	2	varchar(25)	[]	[]	
AZ JOB	3	varchar(25)	[]	[]	
123 MGR	4	int	[]	[]	MUL
HIREDATE	5	date	[]	[]	
123 SAL	6	int	[]	[]	
123 COMM	7	int	[]	[]	
123 DEPTNO	8	int	[]	[]	MUL

Name	Column	Owner	Ref Table	Type	Ref Object
EMP_DEPT_FK	—	EMP	DEPT	FOREIGN KEY	PRIMARY
EMP_EMP_FK	—	EMP	EMP	FOREIGN KEY	PRIMARY

RANGO_SALARIOS

The screenshot shows the MySQL Workbench interface for the SALGRADE table. The 'SALGRADE' tab is selected. The 'Propiedades' tab is active, showing the table name 'SALGRADE', engine 'InnoDB', auto increment '0', charset 'utf8mb4', and collation 'utf8mb4_0900_ai_ci'. The 'Columns' tab is also visible, showing a list of columns with their data types and constraints. The 'Foreign Keys' tab is selected, showing no foreign keys.

Column Name	#	Data Type	Not Null	Auto Increment	Key
123 GRADE	1	int	[v]	[v]	PRI
123 LOSAL	2	int	[]	[]	
123 HISAL	3	int	[]	[]	

3. Con las sentencias SQL adecuadas añade a cada tabla los registros siguientes

INSERT INTO con los valores para cada tabla. Empleados en orden de referencia de la clave foránea recursiva con la misma tabla de MGR,

```

empleados  DEPT  EMP  SALGRADE  * <54.145.167.147> Script-1 X

USE empleados;

● INSERT INTO DEPT (DEPTNO, DNAME, LOC) VALUES
(10, 'ACCOUNTING', 'NEW YORK'),
(20, 'RESEARCH', 'DALLAS'),
(30, 'SALES', 'CHICAGO'),
(40, 'OPERATIONS', 'BOSTON');

● INSERT INTO EMP (ENO, ENAME, JOB, MGR, HIREDATE, SAL, COMM, DEPTNO) VALUES
(7839, 'KING', 'PRESIDENT', NULL, '1981-11-17', 5000, NULL, 10),
(7566, 'JONES', 'MANAGER', 7839, '1981-04-02', 2975, NULL, 20),
(7698, 'BLAKE', 'MANAGER', 7839, '1981-05-01', 2850, NULL, 30),
(7782, 'CLARK', 'MANAGER', 7839, '1981-06-09', 2450, NULL, 10),
(7788, 'SCOTT', 'ANALYST', 7566, '1982-12-09', 3000, NULL, 20),
(7902, 'FORD', 'ANALYST', 7566, '1981-12-03', 3000, NULL, 20),
(7369, 'SMITH', 'CLERK', 7902, '1980-12-17', 800, NULL, 20),
(7934, 'MILLER', 'CLERK', 7782, '1982-01-23', 1300, NULL, 10),
(7876, 'ADAMS', 'CLERK', 7788, '1983-01-12', 1100, NULL, 20),
(7499, 'ALLEN', 'SALESMAN', 7698, '1981-02-20', 1600, 300, 30),
(7521, 'WARD', 'SALESMAN', 7698, '1981-02-22', 1250, 500, 30),
(7654, 'MARTIN', 'SALESMAN', 7698, '1981-10-28', 1250, 1400, 30),
(7844, 'TURNER', 'SALESMAN', 7698, '1981-09-08', 1500, 0, 30),
(7900, 'JAMES', 'CLERK', 7698, '1981-12-03', 950, NULL, 30);

● INSERT INTO SALGRADE (GRADE, LOSAL, HISAL) VALUES
(1, 700, 1200),
(2, 1201, 1400),
(3, 1401, 2000),
(4, 2001, 3000),
(5, 3001, 9999);

```

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4. Realiza las siguientes consultas mostrando también su salida por pantalla.

1. Seleccionar el nº de empleado, salario, comisión, nº de departamento y fecha de la tabla EMP.

```
SELECT ENO,SAL,COMM,DEPTNO,HIREDATE from EMP;
```

	123 ENO	123 SAL	123 COMM	123 DEPTNO	HIREDATE
1	7.369	800	[NULL]	20	1980-12-17
2	7.499	1.600	300	30	1981-02-20
3	7.521	1.250	500	30	1981-02-22
4	7.566	2.975	[NULL]	20	1981-04-02
5	7.654	1.250	1.400	30	1981-10-28
6	7.698	2.850	[NULL]	30	1981-05-01
7	7.782	2.450	[NULL]	10	1981-06-09
8	7.788	3.000	[NULL]	20	1982-12-09
9	7.839	5.000	[NULL]	10	1981-11-17
10	7.844	1.500	0	30	1981-09-08
11	7.876	1.100	[NULL]	20	1983-01-12
12	7.900	950	[NULL]	30	1981-12-03
13	7.902	3.000	[NULL]	20	1981-12-03
14	7.934	1.300	[NULL]	10	1982-01-23

2. Seleccionar todas las columnas de la tabla DEPT.

```
SELECT * from DEPT;
```

	123 DEPTNO	A-Z DNAME	A-Z LOC
1	10	ACCOUNTING	NEW YORK
2	20	RESEARCH	DALLAS
3	30	SALES	CHICAGO
4	40	OPERATIONS	BOSTON

3. Seleccionar los nombres y los empleos de todos los empleados, ordenados por empleo.

```
SELECT ENAME,JOB from EMP ORDER BY JOB;
```

	A-Z ENAME	A-Z JOB
1	SCOTT	ANALYST
2	FORD	ANALYST
3	SMITH	CLERK
4	ADAMS	CLERK
5	JAMES	CLERK
6	MILLER	CLERK
7	JONES	MANAGER
8	BLAKE	MANAGER
9	CLARK	MANAGER
10	KING	PRESIDENT

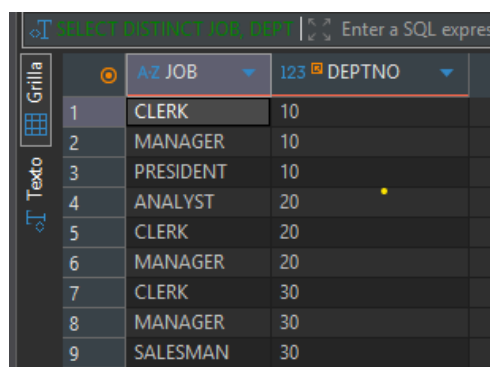
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4. Seleccionar los empleos que hay en cada departamento, ordenados por departamento.

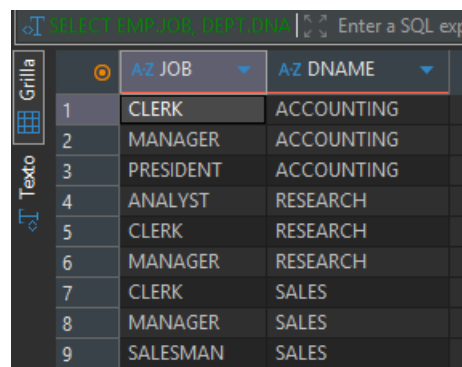
```
SELECT DISTINCT JOB, DEPTNO from EMP ORDER BY DEPTNO;
```

OPCIÓN B (con nombre de departamento)

```
SELECT EMP.JOB, DEPT.DNAME FROM DEPT JOIN EMP ON  
DEPT.DEPTNO = EMP.DEPTNO GROUP BY DEPT.DEPTNO, EMP.JOB  
ORDER BY DEPT.DEPTNO;
```



	AZ JOB	123 DEPTNO
1	CLERK	10
2	MANAGER	10
3	PRESIDENT	10
4	ANALYST	20
5	CLERK	20
6	MANAGER	20
7	CLERK	30
8	MANAGER	30
9	SALESMAN	30



	AZ JOB	AZ DNAME
1	CLERK	ACCOUNTING
2	MANAGER	ACCOUNTING
3	PRESIDENT	ACCOUNTING
4	ANALYST	RESEARCH
5	CLERK	RESEARCH
6	MANAGER	RESEARCH
7	CLERK	SALES
8	MANAGER	SALES
9	SALESMAN	SALES

5. Seleccionar los distintos departamentos que existen en la tabla EMP.

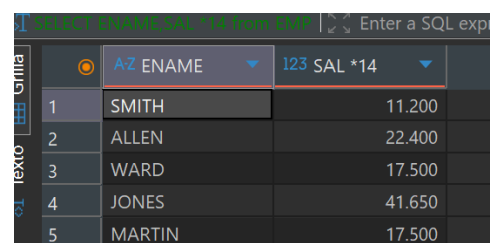
```
SELECT DISTINCT DEPTNO from EMP;
```



	123 DEPTNO
1	10
2	20
3	30

6. Calcular el salario anual a percibir por cada empleado.

```
SELECT ENAME, SAL *14 from EMP;
```



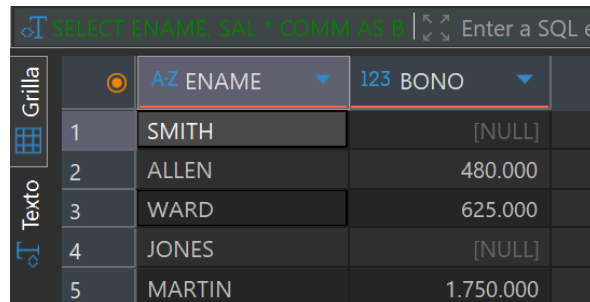
	AZ ENAME	123 SAL *14
1	SMITH	11.200
2	ALLEN	22.400
3	WARD	17.500
4	JONES	41.650
5	MARTIN	17.500

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- Mostrar el nombre del empleado y una columna que contenga el salario multiplicado por la comisión cuya cabecera sea "BONO".

```
SELECT ENAME, SAL * COMM AS BONO from EMP;
```

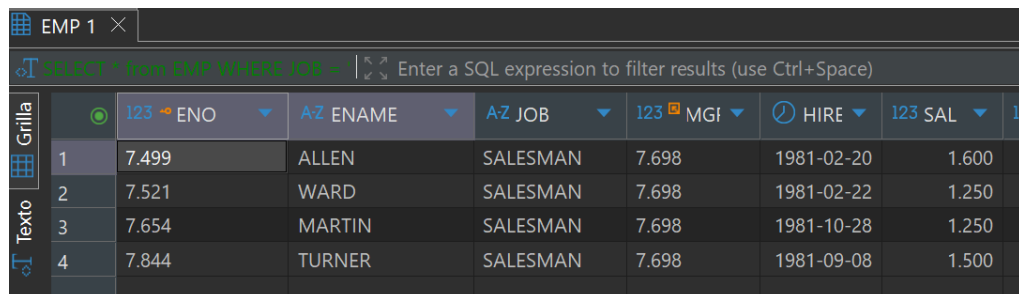
NULL en los campos de comisión que sea NULL



	ENAME	BONO
1	SMITH	[NULL]
2	ALLEN	480.000
3	WARD	625.000
4	JONES	[NULL]
5	MARTIN	1.750.000

- Seleccionar aquellos empleados que sean "SALESMAN".

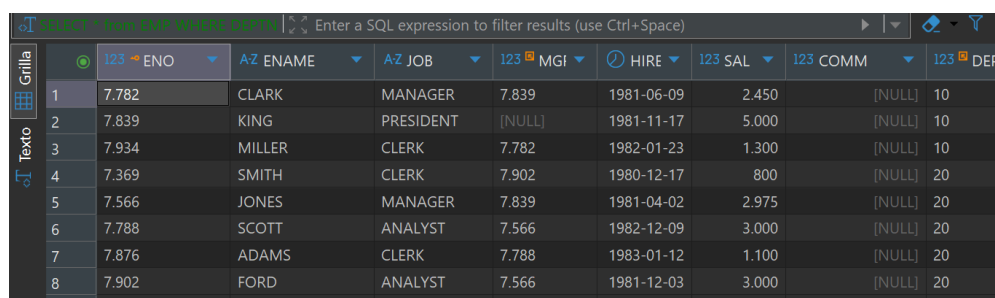
```
SELECT * from EMP WHERE JOB = "SALESMAN";
```



	ENO	ENAME	JOB	MGR	HIRE	SAL	DEPTNO
1	7.499	ALLEN	SALESMAN	7.698	1981-02-20	1.600	10
2	7.521	WARD	SALESMAN	7.698	1981-02-22	1.250	10
3	7.654	MARTIN	SALESMAN	7.698	1981-10-28	1.250	10
4	7.844	TURNER	SALESMAN	7.698	1981-09-08	1.500	10

- Seleccionar aquellos empleados que no trabajen en el departamento 30.

```
SELECT * from EMP WHERE DEPTNO != 30;
```

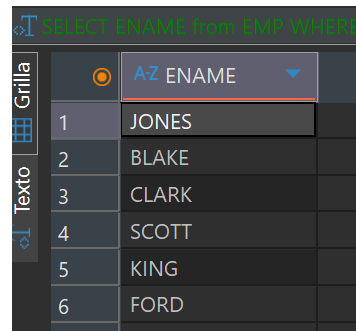


	ENO	ENAME	JOB	MGR	HIRE	SAL	COMM	DEPTNO
1	7.782	CLARK	MANAGER	7.839	1981-06-09	2.450	[NULL]	10
2	7.839	KING	PRESIDENT	[NULL]	1981-11-17	5.000	[NULL]	10
3	7.934	MILLER	CLERK	7.782	1982-01-23	1.300	[NULL]	10
4	7.369	SMITH	CLERK	7.902	1980-12-17	800	[NULL]	20
5	7.566	JONES	MANAGER	7.839	1981-04-02	2.975	[NULL]	20
6	7.788	SCOTT	ANALYST	7.566	1982-12-09	3.000	[NULL]	20
7	7.876	ADAMS	CLERK	7.788	1983-01-12	1.100	[NULL]	20
8	7.902	FORD	ANALYST	7.566	1981-12-03	3.000	[NULL]	20

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10. Seleccionar el nombre de aquellos empleados que ganen más de 2000.

```
SELECT ENAME from EMP WHERE SAL >2000;
```

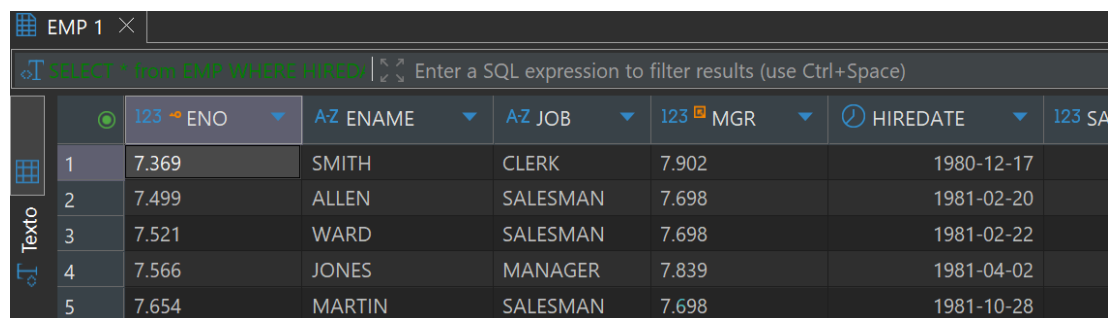


The screenshot shows a database query result in a table view. The query is 'SELECT ENAME from EMP WHERE SAL >2000;'. The table has two columns: 'A-Z ENAME' and an empty column. The data is as follows:

	A-Z ENAME	
1	JONES	
2	BLAKE	
3	CLARK	
4	SCOTT	
5	KING	
6	FORD	

11. Seleccionar aquellos empleados que hayan entrado antes del 1/1/82

```
SELECT * from EMP WHERE HIREDATE <'1982-01-01';
```



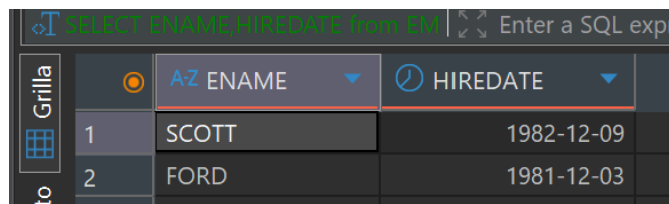
The screenshot shows a database query result in a table view. The query is 'SELECT * from EMP WHERE HIREDATE <'1982-01-01';'. The table has seven columns: '123 ENO', 'A-Z ENAME', 'A-Z JOB', '123 MGR', 'HIREDATE', and '123 SA'. The data is as follows:

	123 ENO	A-Z ENAME	A-Z JOB	123 MGR	HIREDATE	123 SA
1	7.369	SMITH	CLERK	7.902	1980-12-17	
2	7.499	ALLEN	SALESMAN	7.698	1981-02-20	
3	7.521	WARD	SALESMAN	7.698	1981-02-22	
4	7.566	JONES	MANAGER	7.839	1981-04-02	
5	7.654	MARTIN	SALESMAN	7.698	1981-10-28	

12. Mostrar el nombre del empleado y su fecha de alta en la empresa de los empleados que son “ANALISTA”.

* “ANALYST”

```
SELECT ENAME,HIREDATE from EMP WHERE JOB ="ANALYST";
```



The screenshot shows a database query result in a table view. The query is 'SELECT ENAME,HIREDATE from EMP WHERE JOB ="ANALYST";'. The table has two columns: 'A-Z ENAME' and 'HIREDATE'. The data is as follows:

	A-Z ENAME	HIREDATE
1	SCOTT	1982-12-09
2	FORD	1981-12-03

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13. Seleccionar los empleados cuyo salario sea superior al de "ADAMS".

```
SELECT * from EMP WHERE SAL > (SELECT SAL from EMP WHERE ENAME = "ADAMS");
```

SELECT * from EMP WHERE SAL > Enter a SQL expression to filter results (use Ctrl+Space)						
Grilla		123 ENO	A-Z ENAME	A-Z JOB	123 MGR	HIREDAT
Texto	1	7.499	ALLEN	SALESMAN	7.698	1981-02-20
	2	7.521	WARD	SALESMAN	7.698	1981-02-22
	3	7.566	JONES	MANAGER	7.839	1981-04-02
	4	7.654	MARTIN	SALESMAN	7.698	1981-10-28
	5	7.698	BLAKE	MANAGER	7.839	1981-05-01

14. Seleccionar los empleados que trabajan en el mismo departamento que "CLARK".

```
SELECT * from EMP WHERE DEPTNO = (SELECT DEPTNO from EMP WHERE ENAME= "CLARK");
```

SELECT * from EMP WHERE DEPTNO Enter a SQL expression to filter results (use Ctrl+Space)						
Grilla		123 ENO	A-Z ENAME	A-Z JOB	123 M	HIREDAT
Texto	1	7.782	CLARK	MANAGER	7.839	1981-06-09
	2	7.839	KING	PRESIDENT	[NULL]	1981-11-17
	3	7.934	MILLER	CLERK	7.782	1982-01-23

15. Encontrar a los empleados cuyo jefe es "BLAKE".

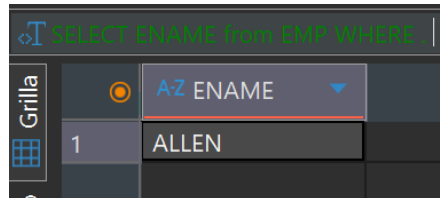
```
SELECT * from EMP WHERE MGR =(SELECT ENO from EMP WHERE ENAME ="BLAKE");
```

SELECT * from EMP WHERE MGR = Enter a SQL expression to filter results (use Ctrl+Space)						
Grilla		123 ENO	A-Z ENAME	A-Z JOB	123 M	HIREDAT
Texto	1	7.499	ALLEN	SALESMAN	7.698	1981-02-20
	2	7.521	WARD	SALESMAN	7.698	1981-02-22
	3	7.654	MARTIN	SALESMAN	7.698	1981-10-28
	4	7.844	TURNER	SALESMAN	7.698	1981-09-08
	5	7.900	JAMES	CLERK	7.698	1981-12-03

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16. Seleccionar el nombre de los vendedores que ganen más de 1500.

```
SELECT ENAME from EMP WHERE JOB= "SALESMAN" AND SAL >1500;
```

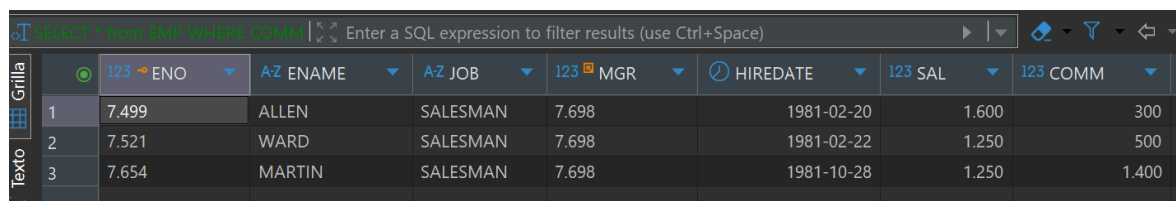


The screenshot shows a SQL query editor with the text "SELECT ENAME from EMP WHERE". Below the editor, a table is displayed with the following data:

Grilla	AZ ENAME
1	ALLEN

17. Seleccionar aquellos empleados que tienen comisión.

```
SELECT * from EMP WHERE COMM IS NOT NULL AND COMM > 0;
```

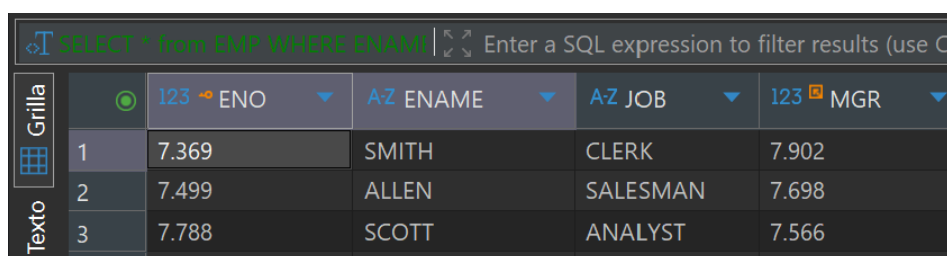


The screenshot shows a SQL query editor with the text "SELECT * from EMP WHERE COMM". Below the editor, a table is displayed with the following data:

Grilla	123 ENO	AZ ENAME	AZ JOB	123 MGR	HIREDATE	123 SAL	123 COMM
1	7.499	ALLEN	SALESMAN	7.698	1981-02-20	1.600	300
2	7.521	WARD	SALESMAN	7.698	1981-02-22	1.250	500
3	7.654	MARTIN	SALESMAN	7.698	1981-10-28	1.250	1.400

18. Seleccionar aquellos que se llamen "SMITH", "ALLEN" o "SCOTT".

```
SELECT * from EMP WHERE ENAME IN ("SMITH", "ALLEN", "SCOTT");
```



The screenshot shows a SQL query editor with the text "SELECT * from EMP WHERE ENAME". Below the editor, a table is displayed with the following data:

Grilla	123 ENO	AZ ENAME	AZ JOB	123 MGR
1	7.369	SMITH	CLERK	7.902
2	7.499	ALLEN	SALESMAN	7.698
3	7.788	SCOTT	ANALYST	7.566

19. Seleccionar aquellos que no se llamen "SMITH", "ALLEN" o "SCOTT".

```
SELECT * from EMP WHERE ENAME NOT IN ("SMITH", "ALLEN", "SCOTT");
```

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SQL: `SELECT * from EMP WHERE ENAME` | Enter a SQL expression to filter results (use Ctrl+Space)

	123 E	A-Z ENA	A-Z JOB	123 MGR
1	7.521	WARD	SALESMAN	7.698
2	7.566	JONES	MANAGER	7.839
3	7.654	MARTIN	SALESMAN	7.698
4	7.698	BLAKE	MANAGER	7.839
5	7.782	CLARK	MANAGER	7.839
6	7.839	KING	PRESIDENT	[NULL]

20. Seleccionar los empleados que trabajen en "CHICAGO".

```
SELECT * from EMP WHERE DEPTNO=(SELECT DEPTNO from DEPT
WHERE LOC="CHICAGO");
```

SQL: `SELECT * from EMP WHERE DEPTNO` | Enter a SQL expression to filter results (use Ctrl+Space)

	123 E	A-Z ENA	A-Z JOB	123 S	123 H	123 S	123 C	123 DEPTNO
1	7.499	ALLEN	SALESMAN	7.698	1981-02-20	1.600	300	30
2	7.521	WARD	SALESMAN	7.698	1981-02-22	1.250	500	30
3	7.654	MARTIN	SALESMAN	7.698	1981-10-28	1.250	1.400	30
4	7.698	BLAKE	MANAGER	7.839	1981-05-01	2.850	[NULL]	30
5	7.844	TURNER	SALESMAN	7.698	1981-09-08	1.500	0	30
6	7.900	JAMES	CLERK	7.698	1981-12-03	950	[NULL]	30

21. Seleccionar aquellos empleados que trabajen en el departamento 10 o en el 20.

```
SELECT * from EMP WHERE DEPTNO=10 OR DEPTNO=20;
```

SQL: `SELECT * from EMP WHERE DEPTNO` | Enter a SQL expression to filter results (use Ctrl+Space)

	123 E	A-Z ENA	A-Z JOB	123 S	123 H	123 S	123 C	123 DEPTNO
1	7.782	CLARK	MANAGER	7.839	1981-06-09	2.450	[NULL]	10
2	7.839	KING	PRESIDENT	[NULL]	1981-11-17	5.000	[NULL]	10
3	7.934	MILLER	CLERK	7.782	1982-01-23	1.300	[NULL]	10
4	7.369	SMITH	CLERK	7.902	1980-12-17	800	[NULL]	20
5	7.566	JONES	MANAGER	7.839	1981-04-02	2.975	[NULL]	20
6	7.788	SCOTT	ANALYST	7.566	1982-12-09	3.000	[NULL]	20
7	7.876	ADAMS	CLERK	7.788	1983-01-12	1.100	[NULL]	20
8	7.902	FORD	ANALYST	7.566	1981-12-03	3.000	[NULL]	20